



Multicentric Analysis of Erythromycin Resistance Determinants in Invasive *Streptococcus pneumoniae*; Associated Serotypes and Sequence Types in India

Rosemol Varghese¹ · Jones Lionel Daniel¹ · Ayyanraj Neeravi¹ · Pavithra Baskar¹ · Anand Manoharan² · Balasubramanian Sundaram³ · Vikas Manchanda⁴ · Karnika Saigal⁵ · Binesh Lal Yesudhasan¹ · Balaji Veeraraghavan¹

Received: 7 December 2020 / Accepted: 25 June 2021 / Published online: 5 July 2021

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Abstract

Streptococcus pneumoniae is the major cause of childhood pneumonia and related deaths in India. Widespread use of erythromycin for the treatment of pneumonia has led to the emergence of erythromycin resistance. Despite this increase in erythromycin resistance, there are very little data on resistance determinants from India. Hence, we aimed to perform the molecular characterization of erythromycin-resistant invasive pneumococcal isolates in India. In this study, 250 erythromycin-resistant invasive isolates obtained from four Indian hospitals between 2014 and 2019 were included. The isolates were reconfirmed by standard CDC protocols, followed by detection of *erm(B)*, *mef(A/E)* genes, and screening for mutations in 23S rRNA, ribosomal proteins L4 and L22. Among the 250 erythromycin-resistant isolates, 46% ($n = 114$) and 35% ($n = 87$) carried the *mef(A/E)* gene and *erm(B)* gene, respectively; both genes were present in 8% ($n = 20$) of the isolates and 12% ($n = 29$) of the studied strains did not bear any of them. The major mutations associated with erythromycin resistance in 23S rRNA, such as A2060C, A2061G, and C2613G, were absent. The predominant serotypes were 19F, 14, 23F, 6A, 6B, 19A, and 9V. The major clonal complexes were CC320, followed by CC230 and CC63. The predominant gene was *mef(A/E)*, and most of the serotypes were PCV13 (54%). This study contributes to the baseline understanding of the erythromycin resistance determinants associated with the serotypes and sequence types (ST) of Indian invasive *S. pneumoniae*.

Introduction

Among the wide range of *Pneumococcal* diseases, *Pneumococcal pneumonia*, which can manifest as invasive or non-invasive, accounts for 36% of overall childhood pneumonia [1]. Farooqi et al. (2015) estimated that in 2010, India witnessed 0.56 million severe episodes of pneumococcal

pneumonia and 105 thousand pneumococcal deaths in children younger than five years of age [2]. Erythromycin was first introduced in 1952 [3], but it was not frequently used for the treatment of pneumonia, until the 1980s. The 1990s marked the rise in the incidence of pneumococcal penicillin resistance [4]. Macrolides, including erythromycin, possess anti-inflammatory and immunomodulatory activity in the eukaryotes [5], making them an optimal drug for upper respiratory infections, especially pneumonia. The increasing incidence of penicillin resistance led to the widespread use of macrolides, which exerted an intense selective pressure that resulted in macrolide-resistant pneumococci [6]. Recent reports show that macrolide resistance among *S. pneumoniae* is geographically variable, ranging from a minimum of 30 to 50% globally [7]. Erythromycin resistance is higher in Asia [8], with a trend of steady progression over the last decade compared to other continents [9, 10]. Until 2004, there were no reports of macrolide resistance from Asia, although several studies from Europe, America, and Africa reported an increasingly high percentage of macrolide resistance [11, 12]. In 2004, Song et al. (ANSORP study) reported

✉ Balaji Veeraraghavan
vbalaji@cmcvellore.ac.in

¹ Department of Clinical Microbiology, Christian Medical College, Vellore 632004, TN, India

² The CHILDS Trust Medical Research Foundation, Chennai, TN, India

³ Kanchi Kamakoti CHILDS Trust Hospital, Chennai, TN, India

⁴ Department of Microbiology, Maulana Azad Medical College, New Delhi, Delhi, India

⁵ Department of Microbiology, Chacha Nehru Bal Chikitsalaya, New Delhi, Delhi, India

a high incidence of macrolide resistance in Asia [11]. In India, macrolide resistance in invasive pneumococcal disease (IPD) among children less than five years increased from 13% in 1999 [13] to 50% in 2019 [14].

Macrolides bind reversibly to the 23S rRNA at a site near the peptidyl transferase, in the center of the 50S ribosomal subunit, where they interfere with the development of peptide bonds during protein elongation, thereby inhibiting protein biosynthesis [15]. The erythromycin-ribosome methylase (*erm*) gene encodes adenine-specific *N*-methyltransferases that confers macrolide resistance by the target site modification. The *erm* gene methylates 23S rRNA and thereby inhibits the binding of macrolide antibiotics [16]. In *S. pneumoniae*, *erm(B)* is the principal ribosomal methylase, whereas the other gene subclasses such as *erm(A)* [17] and *erm(TR)* [18] are rare. The *erm(B)* gene mediates high-level resistance, while the macrolide efflux pump *mef(A/E)* gene mediates low-level resistance. The point mutations in 23S rRNA, L4, or L22 ribosomal proteins can also mediate erythromycin resistance [16]. Although there have been reports of increasing erythromycin resistance in India [19–21], there are very little data on the resistance determinants. Hence, we aimed to perform the molecular characterization of erythromycin-resistant invasive pneumococcal isolates by screening for *erm(B)*, *mef(A/E)* genes and by mutational analysis of both the 23S rRNA and the ribosomal proteins. We also aimed to describe the pneumococcal serotype distribution within the erythromycin-resistant isolates.

Materials and Methods

Bacterial Isolate Characterization and Antimicrobial Susceptibility Testing

This study included 250 archived non-duplicate erythromycin-resistant (blood, cerebrospinal fluid, and other sterile body fluids) invasive pneumococcal isolates in all age groups, collected from July 2014 to December 2019. This laboratory-based study was conducted at Christian Medical College (CMC), Vellore, India, the WHO Pneumococcal reference laboratory in Southeast Asia. The 250 isolates included 200 isolates from CMC, and the remaining 50 isolates from other Indian hospitals located in New Delhi (Chacha Nehru Bal Chikitsalaya, Maulana Azad Medical College), and Chennai (Kanchi Kamakoti CHILDS Trust Hospital). Most of the isolates (76%) were from blood, followed by CSF (14%) and pleural fluid (10%). The isolates from children and adults were 65% and 35%, respectively. The isolates were identified based on the typical

colony morphology, gram staining, optochin sensitivity test (Oxoid Company, Britain), and the bile solubility test. Serotyping was performed by the Co-agglutination method (neufeld antisera obtained from Statens Serum Institut, Copenhagen, Denmark) and customized conventional sequential multiplex PCR according to the Centers for Disease Control and Prevention (CDC) protocol [22]. The minimum inhibitory concentration (MIC) was determined using the Vitek systems II method for erythromycin, and the results were interpreted based on the CLSI (Clinical and Laboratory Standards Institute, M100, 30th edition, 2020) guidelines. ATCC 49619 *S. pneumoniae* strain was the reference for antimicrobial susceptibility testing. For erythromycin, the breakpoints used were ≤ 0.25 , 0.5, and 1 $\mu\text{g/mL}$ for susceptible, intermediate, and resistant, respectively. The MIC value for all the study isolates was more than 1 $\mu\text{g/mL}$.

DNA Extraction, MLST, and PCR

According to the manufacturer's instructions, DNA extracted from an overnight culture grown at 37 °C on blood agar, using QIAamp DNA Mini Kit and the QIA Symphony SP instrument (Qiagen, Hilden, Germany). To detect erythromycin resistance genes *mef(A/E)* and *erm(B)*, PCR was performed on all the isolates using primers described elsewhere [23]. Briefly, the PCR reaction volume consisted of 12.5 μL of master mix, 8.5 μL of nuclease-free H₂O, 2 μL of primer mix, and 2 μL of DNA template, with a total volume of 25 μL . The PCR cycling conditions used were: initial denaturation at 95 °C for 5 min, followed by 35 cycles of denaturation at 94 °C for 1 min, annealing at 48 °C for 1 min, and extension at 72 °C for 1 min and then final extension at 72 °C for 10 min. Among the 250 isolates, 40 were subjected to Multilocus Sequence Typing (MLST) described elsewhere [24]. The sequence type was assigned based on the allelic profile of the seven house-keeping genes using the pubmlst database (https://pubmlst.org/bigdb?db=pubmlst_spneumoniae_seqdef). Nine isolates, negative for the presence of *mef(A/E)/erm(B)*, underwent mutational analysis of 23S rRNA, ribosomal proteins L4 and L22 [23]. The amplified gene products were Sanger sequenced, edited, and merged using the Bioedit free software. The mutational analysis was performed using *S. pneumoniae* R6 as the reference sequence. The amino acid mutations were determined by translation of the DNA sequences using the ExPASy translate tool (<https://web.expasy.org/translate/>) and then compared with the corresponding reference *S. pneumoniae* R6 ribosomal proteins. The alignments were performed using ClustalW sequence alignment software (<http://www.ebi.ac.uk/Tools/msa/clustalo/>).

Results

Detection of Macrolide-Resistant Genes, *erm(B)* and *mef(A/E)*

Among the 250 erythromycin-resistant studied isolates, 46% ($n=114$) and 35% ($n=87$) isolates individually carried *mef(A/E)* gene and *erm(B)* genes, respectively. Both genes, *mef(A/E)* and *erm(B)*, coexisted in 8% ($n=20$) of the isolates, and neither of them were detected in 12% ($n=29$) of the studied isolates.

Serotype Distribution

The predominant serotypes were 19F ($n=31$), 14 ($n=32$), 6A ($n=19$), 6B ($n=26$), 19A ($n=16$), 23F ($n=22$), 9 V ($n=19$), and 38 ($n=8$). The serotype distribution in children and adults is shown in Fig. 1. The expected serotype coverage in children for vaccines PCV13, PCV10SII (Serum Institute of India), PCV10(GSK), PCV15, and PCV20 are 53%, 51%, 43%, 54%, and 57%, respectively. In adults, PPSV23 provides a serotype coverage of 61%. Non-vaccine serotypes 38 (8%) and 7B (4.5%) were predominant among adults. Although there were fewer samples from CSF than from blood, the predominant serotype in CSF was 14, whereas in blood and pleural fluid the predominant serotype was serotype 19F.

MLST

Among the 40 isolates, 29 sequence types were observed within 13 serotypes. Global comparison of studied STs

revealed the grouping of the 29 STs into 13 clonal complexes (CC) and six singletons. The predominant CC was CC320 ($n=13$), followed by CC230 ($n=4$) and CC63 ($n=4$). The PMEN (Pneumococcal Molecular Epidemiology Network) clones associated with these CCs are Taiwan^{19F}-ST236, Spain^{6B-2} ST90, Portugal^{19F}ST177, Sweden^{15A}-ST63, Denmark¹⁴-ST230, and Sweden¹-ST21 (Table 1).

Detection of Mutation in 23S rRNA, L4, and L22

Nine isolates, negative for *erm(B)* and *mef(A/E)* genes, were screened for mutations in both, the 23S rRNA and ribosomal proteins (L4 and L22). All the isolates showed mutations in the first part of the sequence, including domain II, whereas no mutations in the region of domain III–V was found. The mutations observed were A138G, T389C, A682G, C1180T, C1618A, T1745A, A2000C, A2288G, G2519T, G2590T, and A2751T. The earlier reported major mutations associated with erythromycin and clindamycin resistance in 23S rRNA, such as A2060C, A2061G, and C2613G, were absent. None of the isolates showed mutations in L22. The L4 ribosomal protein showed only two mutations, E161G ($n=2$) and S20N ($n=3$), in five isolates, while the conserved site (between the 63 and 74 amino acid positions) had no mutations (Table 2).

Discussion

This study detected the macrolide resistance determinants prevalent among invasive pneumococcal isolates in India. In the present study, *mef(A/E)* was the most prevalent gene, followed by *erm(B)*. These findings are similar to the reports of

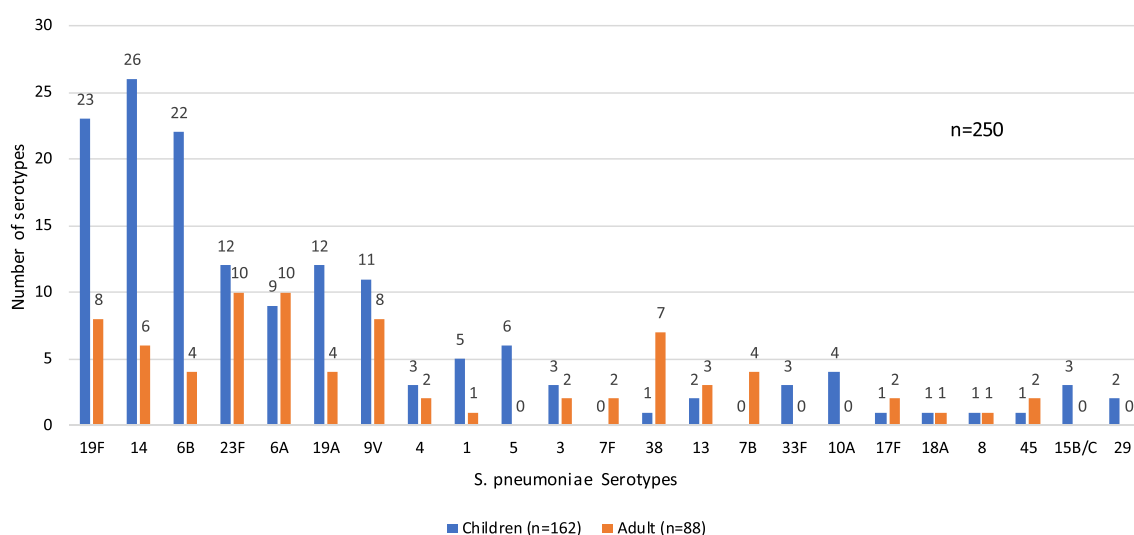


Fig. 1 Serotype distribution among children and adults. Serotypes 2, 34, 36, 23A, 24F, 6C, 39, 10C, 15A, 22A, 35A, 6D, 24F, and 25F were of one each

Table 1 Serotype, sequence type, resistant genes, and erythromycin MIC values for study isolates

S.No	Isolate ID	Serotype	Erythromycin MIC(μg/ml)	<i>Mef(A/E)</i> gene	<i>erm(B)</i> gene	ST	CC
1	B4733	19F	4	<i>Mef(A/E)</i>	Neg	236	320
2	B6024	6A	2	<i>Mef(A/E)</i>	Neg	473	473
3	B11378	19F	4	<i>Mef(A/E)</i>	Neg	236	320
4	B44415	19A	8	<i>Mef(A/E)</i>	<i>erm(B)</i>	320	320
5	B901	6B	1	Neg	Neg	320	320
6	B2393	1	1	Neg	Neg	303	217
7	B2994	14	1	Neg	Neg	13,055	S
8	B7396	6B	1	Neg	<i>erm(B)</i>	9432	S
9	B7765	19F	1	<i>Mef(A/E)</i>	Neg	9428	S
10	B11342	23F	1	Neg	Neg	9424	S
11	B23021	6B	1	<i>Mef(A/E)</i>	<i>erm(B)</i>	9430	S
12	B30300	8	1	Neg	Neg	6197	6197
13	B33430	6A	1	Neg	Neg	9431	63
14	B33586	18A	1	Neg	Neg	63	63
15	B35841	6B	1	Neg	<i>erm(B)</i>	9434	S
16	B21564	19F	1	Neg	<i>erm(B)</i>	13,072	320
17	B36882	6B	> 8	<i>Mef(A/E)</i>	Neg	271	320
18	BVM3	19 F	1	<i>Mef(A/E)</i>	<i>erm(B)</i>	271	320
19	BVM5	6B	1	Neg	<i>erm(B)</i>	874	874
20	B373	10A	2	Neg	Neg	3135	230
21	B376	10A	8	Neg	Neg	3135	230
22	B383	10A	8	<i>Mef(A/E)</i>	Neg	3135	230
23	B471	14	2	<i>Mef(A/E)</i>	Neg	4532	63
24	BVM1402	4	8	Neg	<i>erm(B)</i>	5872	5872
25	B20479	6B	8	Neg	<i>erm(B)</i>	90	90
26	B20490	6B	8	Neg	<i>erm(B)</i>	13,078	90
27	B20527	23F	8	Neg	Neg	10,812	10,812
28	BC1401	45	1	Neg	<i>erm(B)</i>	2212	2212
29	B2352	NT	1	<i>Mef(A/E)</i>	Neg	13,160	320
30	BH123	6A	1	Neg	<i>erm(B)</i>	13,079	473
31	CS341	14	8	<i>Mef(A/E)</i>	<i>erm(B)</i>	236	320
32	BP1024	23F	8	Neg	<i>erm(B)</i>	1701	230
33	BP15252	19F	8	<i>Mef(A/E)</i>	<i>erm(B)</i>	320	320
34	FF178	19A	8	<i>Mef(A/E)</i>	<i>erm(B)</i>	320	320
35	CS2240	19F	8	<i>Mef(A/E)</i>	<i>erm(B)</i>	271	320
36	BA13858	4	8	Neg	<i>erm(B)</i>	14,641	15,688
37	BP10735	14	8	Neg	<i>erm(B)</i>	63	63
38	B1010	6B/E	8	Neg	<i>erm(B)</i>	874	874
39	B2010	19F	8	<i>Mef(A/E)</i>	<i>erm(B)</i>	2812	320
40	FF3287	19F	8	Neg	Neg	179	177

ST sequence type, CC clonal complex, S Singleton, Neg negative

CCs in bold denotes related PMEN clones

the ANSORP (Asian Network for Surveillance of Resistant Pathogens) (12) study, and contradictory to the findings by Peela et al., where *erm(B)* was predominant (20).

India, compared to other countries, such as South Africa, Australia, Turkey, and the USA [25–28], has reported a low percentage of isolates bearing both *mef(A/E)* and *erm(B)*

genes. Isolates harboring dual macrolide resistance determinants are associated with the genetic elements of multi-drug-resistant clonal complexes; hence, they are known to display resistance to multiple classes of antimicrobial agents [29]. The STs of the studied isolates having both *erm(B)* and *mef(A/E)* genes were predominantly from CC320 and

Table 2 Mutations observed in the 23S rRNA, ribosomal proteins L4 and L22

Lab no	ST	CC	Serotype	23S rRNA (2902 bp)	Mutations in 23S rRNA domain V 2042-2628				Mutations in	
					I (23')	II (18')	III (30')	IV (91')	L4	L22
B2393	303	217	1	A50G, G107T, G260A, T389C, A682G, C1618A	None	None	None	None	E161G	None
B33430	9431	63	6A	A138G, T389C, A682G, C1618A, T1745A	A2000C, A2288G, G2519T, G2590T, A2751T	NA	None	None	None	None
B2994	13,055	Singleton	14	G260A, T389C, A682G, C1618A, T1745A	None	NA	None	None	S20N	None
B30300	6197	10,588	8	A138G, G260A, T389C, A682G, C1180T, C1618A, T 1745A	None	NA	None	None	None	None
B33586	63	63	18A	G260A, T389C, A682G, C1618A	A2000C	None	NA	None	None	None
B901	13,057	320	6B	G260A, T389C, A682G, A1140G, C1618A, A2136G	NA	None	T2099C	E161G	None	None
B373	3135	230	10A	G150T, G260A, A682G, C1618A, T1745A	NA	NA	NA	S20N	None	None
B 376	3135	230	10A	G150T, G260A, A682, C1618A, T1745A	NA	NA	NA	S20N	None	None
B20527	10,812	5654	23F	G260A, T389C, A682G, C1618A	None	None	None	None	None	None

None—No mutations observed

NA not amplified

have evolved from the Taiwan^{19F}-14 (ST236) clone, which initially had *mef(A/E)* and then later acquired *erm(B)* [30] genes. In 2004, ST236 and ST81 were the most significant erythromycin-resistant lineages reported in Asia, but the study included only one isolate from India [11]. In 2011, ST320 replaced ST236 and ST81 as the predominant lineage of erythromycin resistance in Asia [31]. However, this study involved many isolates collected from Korea and very few from India. The current study is the first to include STs of erythromycin-resistant invasive pneumococcal isolates from India.

Observation of similar penicillin MIC value of the *S. pneumoniae* isolates of a clonal complex to the associated individual PMEN clones explains the gradual evolution of these clones by recombination [32]. Therefore, there is clonality observed within the sequences of penicillin binding proteins (PBPs). While erythromycin resistance in *S. pneumoniae* is mainly due to mobile genetic elements, the clonality seen in penicillin resistance is less seen in erythromycin-resistant clones and ensures the rapid spread of resistance. In this study, the major mutations in the 23S rRNA domain V responsible for high erythromycin resistance observed in other studies, such as A2058G, C2611G (MLS_B), A2059G, and C2610G, were not detected. In the current study, mutations in the 23S rRNA (A138G, G260A, T389C, A682G, C1180T, C1618A, T 1745A) were within the first 2000 bp, which encodes for the domain I and II, and no mutations in domain V of the 23S rRNA were observed. Of these, A138G and T389C have been reported previously. This result is in concordance with

the findings of Canu et al. on telithromycin resistant *S. pneumoniae* [33]. Mutations in the domain II and IV of 23S rRNA mainly affects the macrolide resistance in *S. pneumoniae* [33, 34]. Among the L4 ribosomal protein mutations, only S20N has been reported earlier from Germany [35]. The absence of major ribosomal mutations indicates decreased spread of resistance mediated by ribosomal mutations. The mobile genetic elements are the predominant mode of erythromycin resistance in Indian isolates of *S. pneumoniae*.

The most prevalent resistant serotypes were of PCV 13 (77%). The recently introduced PCV13 could protect against invasive disease and reduce antimicrobial resistance among the vaccine serotypes in India. Serotype 14 (13%, Fig. 1) was the major serotype, followed by 19F, the same as previous reports from India [14, 21, 28]. These two major serotypes correlate with the predominant CCs associated serotypes: CC320, which is associated with Taiwan^{19F}-ST236 and Denmark¹⁴-ST230 clones. Recent reports of change in the serotypes, antimicrobial susceptibility, and clonality associated with the introduction of pneumococcal childhood vaccination [36–38] mean that it will be imperative to monitor the emergence of non-vaccine serotypes with resistance due to vaccine pressure.

Conclusion

This study contributes to a better understanding of the baseline resistance determinants associated with the serotypes and sequence types of Indian invasive *S. pneumoniae*

resistant to erythromycin. This study reports the predominance of *mef(A/E)* gene-mediated resistance rather than the ribosomal mutations, with the majority being PCV13 serotypes. The rapid spread of erythromycin resistance mediated by mobile genetic elements highlighted the need to discontinue the misuse of macrolides to treat upper respiratory tract infections. The presence of erythromycin-resistant non-vaccine serotypes demands monitoring the prevalent serotypes and their antimicrobial resistance profiles continuously.

Authors Contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by RV, AN, JLK, and PB. RV wrote the first draft of the manuscript and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding No funding was received toward this study.

Declarations

Conflict of interest All the authors declare that they have no conflict of interest.

Ethical Approval The institutional ethical committee of Christian Medical College has approved the study. (Institutional Review Board Min.No:8200).

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